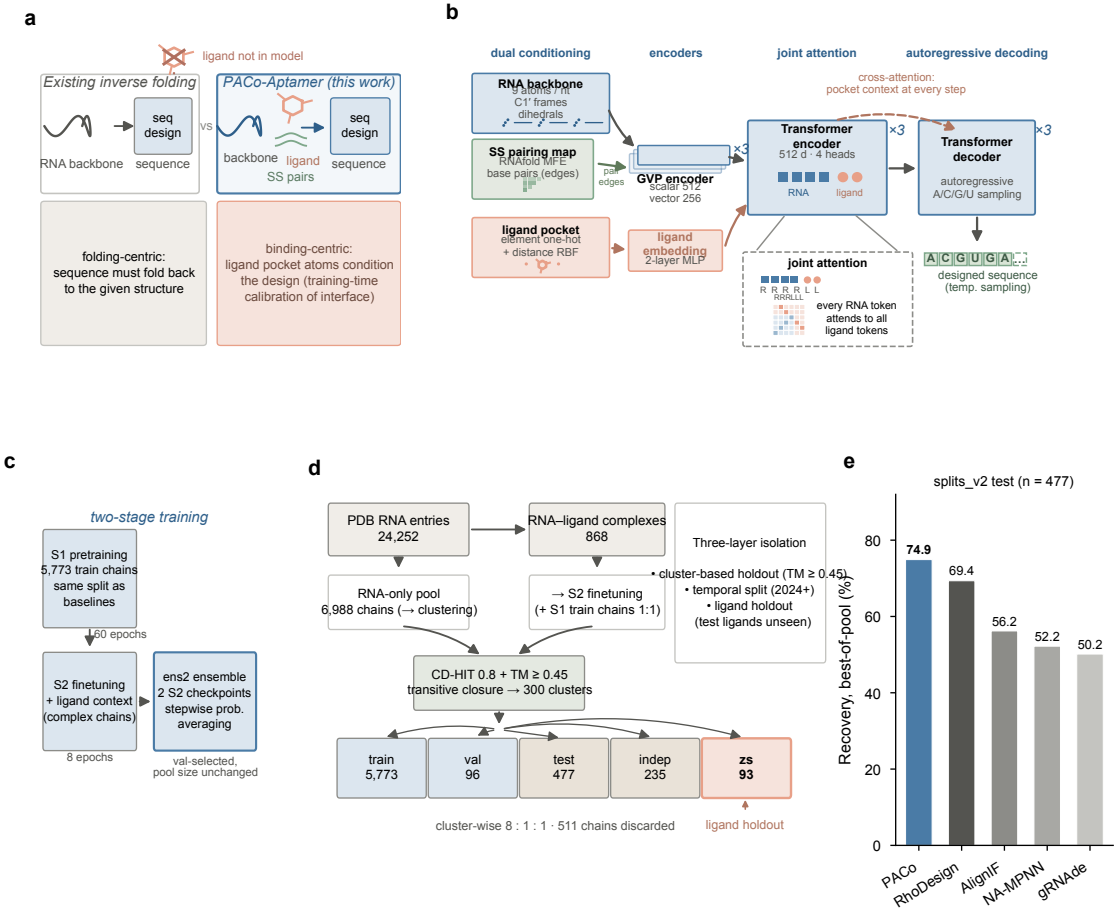

a
ligand not in model
Existing inverse folding
seq design
RNA backbone
sequence
vs
PACo-Aptamer (this work)
seq design
backbone
ligand
SS pairs
sequence
folding-centric: sequence must fold back to the given structure
binding-centric: ligand pocket atoms condition the design (training-time calibration of interface)
b
dual conditioning
encoders
joint attention
autoregressive decoding
RNA backbone
8 atoms / nt
C1′ frames
dihedrals
SS pairing map
RNAfold MFE
base pairs (edges)
pair edges
GVP encoder
scalar 512
vector 256
×3
ligand pocket
element one-hot
+ distance RBF
ligand embedding
2-layer MLP
Transformer encoder
512 d · 4 heads
RNA
ligand
cross-attention:
pocket context at every step
Transformer decoder
autoregressive
A/C/G/U sampling
joint attention
every RNA token attends to all ligand tokens
designed sequence
(temp. sampling)
c
two-stage training
S1 pretraining
5,773 train chains
same split as baselines
60 epochs
S2 finetuning
+ ligand context
(complex chains)
8 epochs
ens2 ensemble
2 S2 checkpoints
stepwise prob. averaging
val-selected, pool size unchanged
d
PDB RNA entries
24,252
RNA–ligand complexes
868
Three-layer isolation
• cluster-based holdout (TM ≥ 0.45)
• temporal split (2024+)
• ligand holdout
(test ligands unseen)
RNA-only pool
6,988 chains (→ clustering)
→ S2 finetuning
(+ S1 train chains 1:1)
CD-HIT 0.8 + TM ≥ 0.45
transitive closure → 300 clusters
train
5,773
val
96
test
477
indep
235
zs
93
cluster-wise 8 : 1 : 1 · 511 chains discarded
ligand holdout
e
splits_v2 test (n = 477)
Recovery, best-of-pool (%)
0
20
40
60
80
74.9
69.4
56.2
52.2
50.2
PACo
RhoDesign
AlignIF
NA-MPNN
gRNAde